

F24-037-D-ArchiVisionAI

Project Team

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Chapter 1

Introduction

Identify the product or application whose requirements are specified in this document. Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. [2].

1.1 Problem Statement

Many widely used softwares either support interior designing like RoomGPT [1] or floor-plan designing like smartDraw [3] causing the user to constantly switch between multiple platforms to enjoy both of these services which is time taking and inefficient. Furthermore, many softwares are not user friendly I.e. the user has to dedicate a lot of time to learn the way the software before using it. Most of the things have to be done manually by dragging and dropping items wasting more of the time. Most of the apps like RoomGPT lacks Augmented Reality (AR) feature and the apps which offer this doesn't provide the option to compare two objects to help the user in selection process. ArchiVision AI solves the problem of switching between multiple apps by integrating architectural design with interior design. It makes the app more user friendly by voice and text commands. For making selection process easy for users it allows comparing multiple object together using AR.

1.2 Scope

ArchiVisionAI is an application that merges architectural design (floorplan generation) with interior design and Augmented Reality (AR). This application is for the ease of people with low or no knowledge of architectural and interior design , not for professional architects specifically . For further elaboration, the floor plans generated will not be in

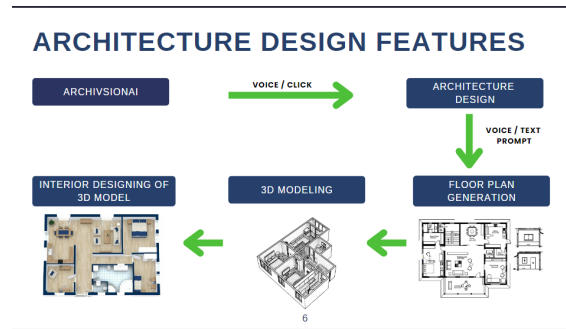


Figure 1.1: Architecture Design

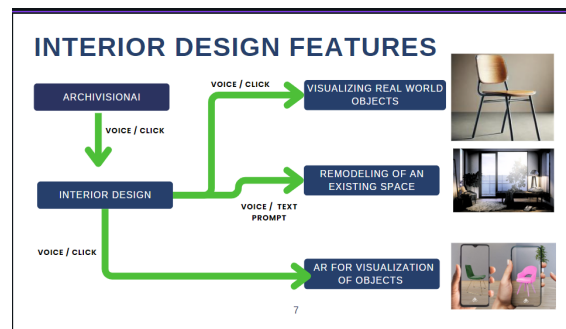


Figure 1.2: Interior Design Diagram

accordance to location , ventilation and other factors which a real architect considers. The sole purpose of this application is to help users visualize and see their living spaces by giving them 3D visualizations of floorplans and interior design options. The user can get a floorplan, visualize it in 3D and decorate it according to his wishes. This application will make decision making more easy with Augmented Reality for visualizing a non existent object in a room or by visualizing an actual object in the picture of a room. The following diagrams show the scope and functionality of our project.

1.3 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

1.3.1 Module 1: Architecture Design Implementation

This module covers all aspects of architectural design within the application, including front-end development.

1. Data Collection and Model Exploration
2. Model Training and Tuning
3. Floor Plan Generation in 2D
4. 3D Plan generation
5. Front-End Design
6. Save Progress
7. Integration of All Features

1.3.2 Module 2: Interior Design Implementation

Interior Design Implementation This module covers all aspects of interior design within the application, including front-end development.

1. Data Collection and Model Exploration
2. Model Training and Tuning
3. Image Capture/Upload Feature
4. Augmented Reality for Viewing Objects in User's Image
5. Smart Purchase Preview (Adding real time objects to an image)
6. Front-End Design
7. Voice based interface inclusion
8. Integration with the Architecture Design Module

1.4 User Classes and Characteristics

User class	Description
Interior Designers	The interior designers are the people who will renovate the houses of people based on the designs ArchiVisionAI will generate. They will also use this system to help the customers understand better about the end results of renovation process. This can also help interior designers to pick a good piece of furniture for users.
Land Owners	Land and owners are people who own a piece of land and want to see how their house will look based on the floorplan or generate a floor plan and visualize it.
House Owners	House owners are people who have their own houses and wish to renovate their rooms or entire house, they can use ArchiVisionAI to create the remodelled version of their houses.
Furniture Businesses	These businesses sell furnitures to people, they can enhance their customers' decision making by using AR to see how a particular furniture i.e, sofas or chairs will look inside customers' rooms.

Chapter 2

Project Requirements

2.1 Use-cases

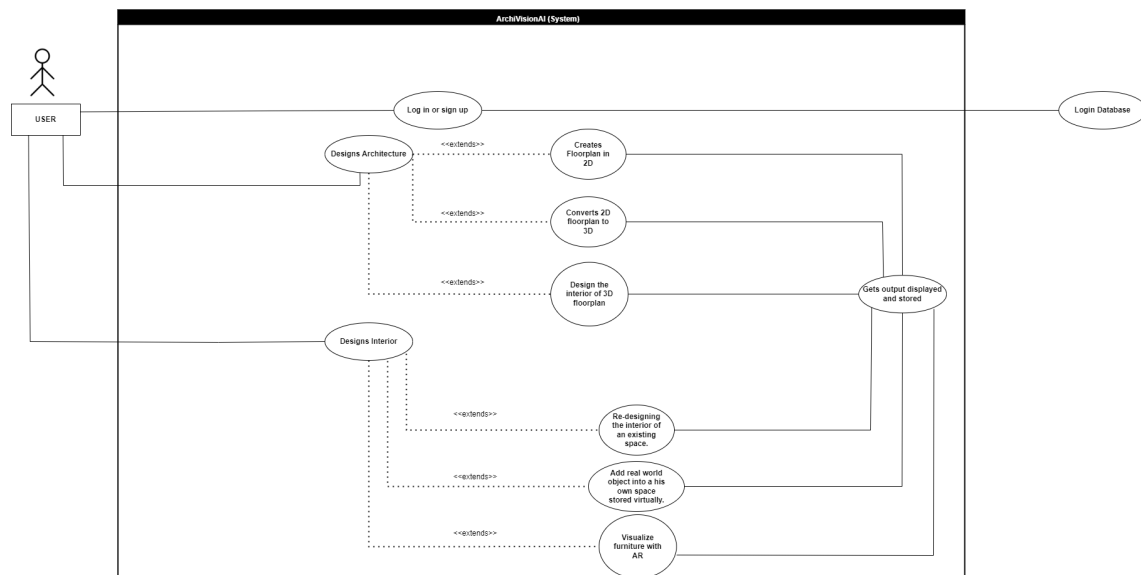


Figure 2.1: Use Case Diagram

2.1.1 Use Cases For Architecture Design Part

2.1.1.1 User Authentication (Sign Up/Log In)

Actor	User
Description	The user can create a new account or sign in to an existing account.
Preconditions	The user needs to authenticate their identity.
Main Flow	1. User selects either "Sign Up" or "Log In." 2. The system asks the user to enter the credentials i.e. Username and Password. 3. User enters Username and Password which are then verified by the system. 4. After successful verification, the user can access and use the application's features.
Postconditions	User is authenticated and granted access to the application.

2.1.1.2 Floor Plan Generation

Actor	User
Description	The user generates a new floor plan based on input that can be a text or voice prompt specifying parameters such as the number of rooms.
Preconditions	The user is logged in and has chosen architecture design followed by floor plan generation.
Main Flow	1. User inputs number of rooms and descriptions of floor plans. 2. The system generates a 2D floor plan. 3. User can review or generate it and convert it to a 3D model.
Postconditions	A 2D floor plan is created and saved for 3D rendering.

2.1.1.3 Floor Plan 3D Model

Actor	User
Description	Converts the generated floor plan into a 3D model for visualization.
Preconditions	A 2D floor plan has been generated or uploaded.
Main Flow	1. User selects the option to visualize the floor plan in 3D. 2. The system renders the 3D model based on the floor plan's specifications. 3. User can design the interior.
Postconditions	A 3D version of the floor plan is generated and its interior can be designed.

2.1.1.4 Interior Design of 3D Model

Actor	User
Description	The user furnishes and designs the interior of the 3D model with AI.
Preconditions	A 3D model of the floor plan exists.
Main Flow	1. User gives prompts for styling the interior. 2. The system updates the 3D model with the design generated by AI.
Postconditions	The interior design of 3D model is saved.

2.1.2 Use Cases For Interior Design Part

2.1.2.1 AR for Viewing Objects

Actor	User
Description	The user visualizes furniture items in real time using Augmented Reality (AR).
Preconditions	The user's device is AR-enabled i.e a smartphone is required.
Main Flow	1. User selects an item from the interior design catalog. 2. The system activates the AR feature, overlaying the item in the real-world space.
Postconditions	The user can visualize and compare items in their own environment.

2.1.2.2 Remodeling Interior

Actor	User
Description	The user modifies the interior design by giving prompts for styling.
Preconditions	The images of the space are uploaded or a 3D model is created.
Main Flow	1. User gives prompts for styling the interior. 2. The system, using AI, generates designs according to the prompts given by the user. 3. User confirms the changes, and the model is updated.
Postconditions	The interior design is remodeled according to the user's preferences.

2.1.2.3 Visualizing Real-World Objects

Actor	User
Description	The user visualizes how existing real-world objects (e.g., new furniture) will look in their space using this feature.
Preconditions	The user's device ensures that the real-world object is scanned. The user's space/environment must be saved in the application or uploaded.
Main Flow	1. User scans an object to visualize. 2. The system overlays the object into the space already uploaded or saved in the application.
Postconditions	The object visualization in user's space is completed, allowing the user to make decisions based on the preview.

2.2 Extended Use-cases

2.2.1 Extended Use-cases for Architectural Design Part

2.2.1.1 User Authentication (Sign Up/Log In)

Use Case ID	UC01
Description	The user either creates a new account or logs into an existing account.
Trigger	The user either selects "Sign Up" or "Log In."
Preconditions	The user has the application installed and an internet connection.
Postconditions	The user is authenticated and granted access to the application's features.
Normal Flow	1. The user clicks on either "Sign Up" or "Log In." 2. The system prompts for credentials i.e. Username and Password. 3. The user enters their username and password. 4. The system validates the credentials. 5. The user is logged in if credentials are correct.
Alternative Flow	If the user forgets their password, the system provides a "Forgot Password" link.
Exception	Incorrect credentials trigger an error message prompting the user to retry.
Frequency of Use	Frequently used when accessing the application.
Special Requirement	Secure password storage and encryption.
Assumptions	Users have valid unique usernames or emails.

2.2.1.2 Floor Plan Generation

Use Case ID	UC02
Description	The user generates a new floor plan based on input prompts with parameters.
Trigger	The user inputs room dimensions, number of rooms, and other specifications.
Preconditions	The user is logged in and ready to start the floor plan by choosing it under the architecture design part.
Postconditions	A 2D floor plan is generated and available for conversion to 3D.
Normal Flow	1. The user inputs prompt having room size, number of rooms, and so on. 2. The system processes these inputs to generate a 2D floor plan. 3. The user reviews and converts it to 3D if needed. 4. The system saves the floor plan.
Alternative Flow	If the user cancels the input process, the system will discard the plan.
Exception	Invalid entries for inputs trigger a warning for the user to correct the inputs.
Frequency of Use	Used whenever the user starts a new architecture design project.
Special Requirement	Precision in the generation of designs.
Assumptions	The user understands how to input prompts with parameters correctly.

2.2.1.3 Floor Plan 3D Model Generation

Use Case ID	UC03
Description	Converts a generated 2D floor plan into a 3D model.
Trigger	The user selects the option to visualize the floor plan in 3D.
Preconditions	A 2D floor plan has been created or uploaded.
Postconditions	The 3D floor plan is visualized.
Normal Flow	1. The user selects "Convert to 3D." 2. The system renders the 3D model. 3. The system saves the model.
Alternative Flow	If rendering fails, the system notifies the user and suggests retrying.
Exception	Complex plans might take longer to render, and the system notifies the user about the delay.
Frequency of Use	Used frequently after creating 2D floor plans.
Special Requirement	High graphical processing capability.
Assumptions	The user's device has enough memory to support 3D rendering.

2.2.1.4 Interior Design of 3D Model

Use Case ID	UC04
Description	The user furnishes and designs the interior of the 3D model with AI.
Trigger	The user selects the option to begin interior design after the 3D model is created.
Preconditions	A 3D model exists, and the user wants to design its interior.
Postconditions	The 3D model is updated with user-preferred designs.
Normal Flow	1. The user opens the 3D model in the interior design module. 2. The user prompts the system for styling the space. 3. The system updates and saves the model with the user's preferred styles.
Alternative Flow	If the user doesn't like the options, the system will regenerate the designs.
Exception	None.
Frequency of Use	Frequently used after 3D model generation.
Special Requirement	AI model should be able to provide diversity in designs.
Assumptions	The user understands how to generate designs using prompts correctly.

2.2.2 Extended Use-cases for Interior Design Part

2.2.2.1 AR for Viewing Objects

Use Case ID	UC05
Description	Users visualize furniture items using Augmented Reality (AR).
Trigger	The user selects an item to preview in AR.
Preconditions	The device supports AR functionality.
Postconditions	The selected item is visualized in the real-world environment via AR.
Normal Flow	1. The user selects an item from the catalog. 2. The system activates the device's AR mode. 3. The user points the device's camera at their space. 4. The system overlays the item into the space.
Alternative Flow	If AR is not supported, the system will suggest using a 3D preview instead.
Exception	Poor lighting or room conditions prevent the AR from functioning correctly.
Frequency of Use	Occasionally used during the interior design process.
Special Requirement	Good quality camera and AR capabilities like in a smartphone.
Assumptions	The user's space is well-lit and suitable for AR visualization.

2.2.2.2 Remodeling Interior Spaces

Use Case ID	UC06
Description	The user modifies the interior design by giving prompts for styling.
Trigger	The user selects an option to remodel an existing space after uploading its images or creating its 3D model.
Preconditions	The user must be logged in, and images or 3D model of the space must exist.
Postconditions	The updated model is saved with the remodeled design changes.
Normal Flow	1. The user gives prompts for styling the interior. 2. The system, using AI, generates designs according to the user's prompts. 3. The user confirms the changes, and the model is updated.
Alternative Flow	If the user doesn't like the designs, the system can regenerate them.
Exception	If a space has not been created or uploaded, it prompts the user to upload an image or create a 3D model.
Frequency of Use	Frequently used when making design adjustments to an existing space.
Special Requirement	The system should support high-resolution rendering and provide diversity in results.
Assumptions	The user is familiar with the correct way to give prompts to the model.

2.2.2.3 Visualizing Real-World Objects in Virtual Space

Use Case ID	UC07
Description	The user scans real-world objects (such as furniture) to visualize them in a virtual environment, allowing comparison with other items or spaces.
Trigger	The user initiates the option to scan a real-world object for visualization.
Preconditions	The user's device must have a camera and be capable of scanning objects. The 3D model or images of the user's space where the object is to be visualized must exist.
Postconditions	The scanned object is accurately represented in the virtual space, allowing the user to decide if they want to purchase it.
Normal Flow	1. The user selects the "Scan Object" option. 2. The system activates the camera and guides the user through the scanning process. 3. The system processes the scan and adds the object to the user's space.
Alternative Flow	If the scanning fails, the system prompts the user to retry or adjust lighting/environment conditions for better accuracy.
Exception	Poor scan quality due to low light or reflective surfaces may result in inaccurate models, and the system will notify the user to improve the scanning conditions.
Frequency of Use	Occasionally used when comparing new furniture or objects with existing spaces.
Special Requirement	The device must have a camera to scan the objects.
Assumptions	The user understands how to scan objects.

2.3 Functional Requirements

The functional requirements of ArchiVisionAI are listed below.

2.3.1 Module 1: Architectural Design

2.3.1.1 User Registration (Login and Signup)

- Users can create an account and access it later with their username and password.

2.3.1.2 2 Dimensional Floor Plan Generation

- Users can generate a floor plan according to the prompts they enter.
- Options to edit the floor plan would also be available.

2.3.1.3 3D Model Generation

- Users can make a 3D model of their 2D floor plan.
- Users have the option to choose their desired layout.

2.3.1.4 User Interface (UI/UX)

- A user-friendly interface for architecture design is present to make full use of the application.
- Navigation between different features is easily accessible.

2.3.2 Module 2: Interior Design

2.3.2.1 Interior Design Model

- The users get interior designs on the basis of their selected themes.
- The user can explore multiple interior design options.

2.3.2.2 Image Capture/Upload

- The users capture and upload images of their spaces to see different design options.
- The users can renovate their existing living spaces.

2.3.2.3 AR for Viewing Limited Objects in Real Time

- The user visualizes furniture items in their spaces through augmented reality in real-time to see the suitable place of an item.
- The operation of AR is smooth for users.

2.3.2.4 Smart Purchase Preview

- The users can add the image of an item in the image of their existing space.
- The users can compare multiple options.

2.3.2.5 User Interface (UI/UX)

- A user-friendly interface for interior designing is present to make full use of the application.
- Navigation between different features is easily accessible.

2.3.2.6 Voice-Based Prompts

- The users have the option to operate the app using voice-based prompts along with text.
- The voice recognition system accurately understands the user.

2.4 Non-Functional Requirements

2.4.1 Reliability

The application should have a mean time between failures (MTBF) of at least 500 hours. Failure is an event where users cannot use main functionalities like 3D modeling, AR visualization, or interior design options. The system will be able to recover from the failure in 15 minutes.

2.4.2 Usability

The application has easy to understand interface which allows users to effortlessly generate floor plans, view furniture in AR, and receive design suggestions with minimal learning time. Users would be able to learn how to use the application within 15 minutes. voice based prompt feature will make usability more easier even for non technical users. A user error rate of less than 10 percent is expected. Example:

- USE-1: The user can change the interior design styling for the same model multiple times.
- USE-2: The user can preview the furniture item using AR within 1 minute after selecting the item.
- USE-3: The user can take a picture of their room . see how an item would look in their room.
- USE-4: The user can see the different interior design styles.
- USE-5: The user can use voice commands to request different design ideas to remodel a specific room (e.g., "Show me modern living room").

2.4.3 Performance

The application should have a maximum response time of 1 minute to get AR visualization and 10 seconds for user interactions like generating a floor plan. For larger and complex tasks like creating a 3D model or running interior design suggestions, the task should not take longer than 15 seconds. Example:

- PER-1: 85 percent of user interactions should meet the 10 second response time.

- PER-2: 90 percent of complex tasks should meet the 15 seconds response time target.
- PER-3: 90 percent of AR tasks should meet the 1 minute response time target.

2.4.4 Security

The system must ensure the protection of sensitive data, prevent unauthorized people from entering the system. The main aim will be safeguarding user information and overall integrity of the application.

- SEC-1: The system will ensure that all user data, including personal details like passwords is kept safe from unauthorized access.
- SEC-2: The application will protect from brute force attacks by limiting the number of login attempts to 5.

2.5 Domain Model

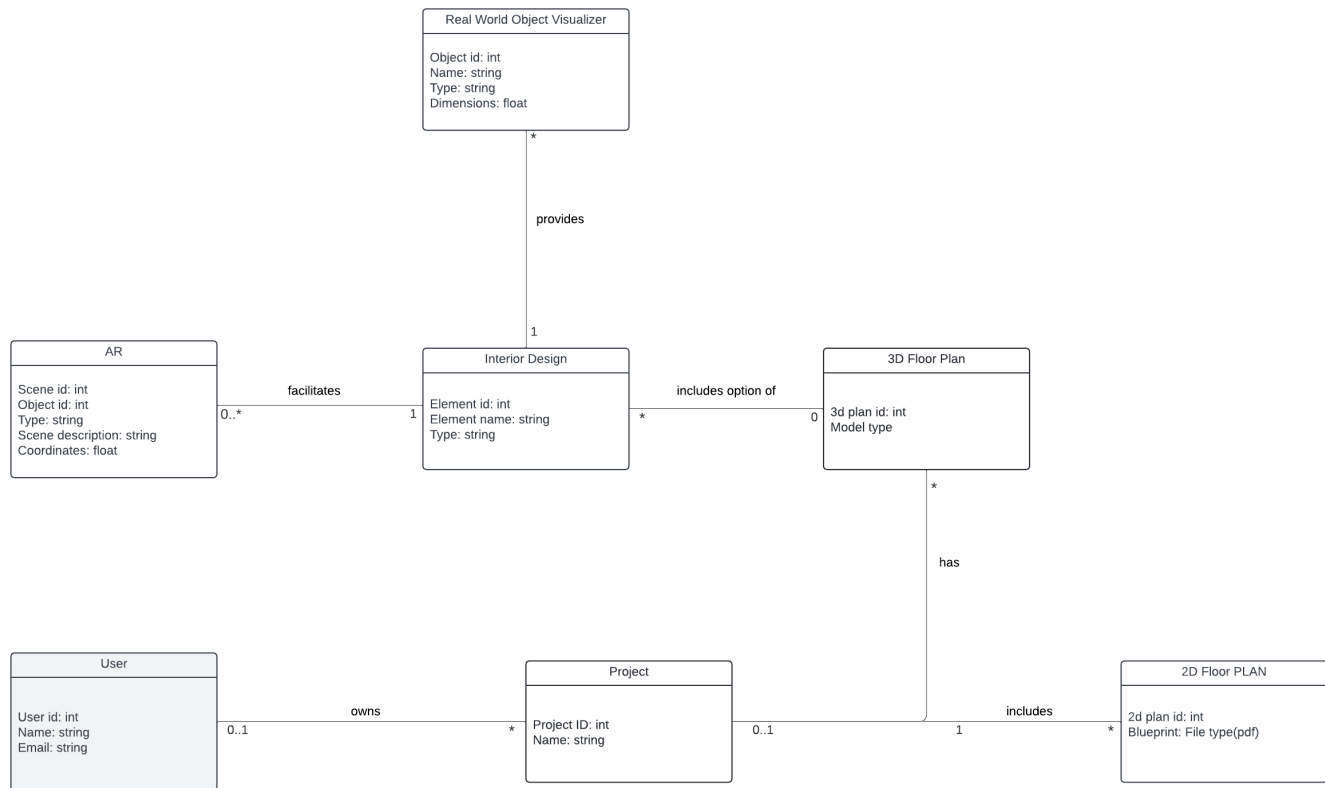


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

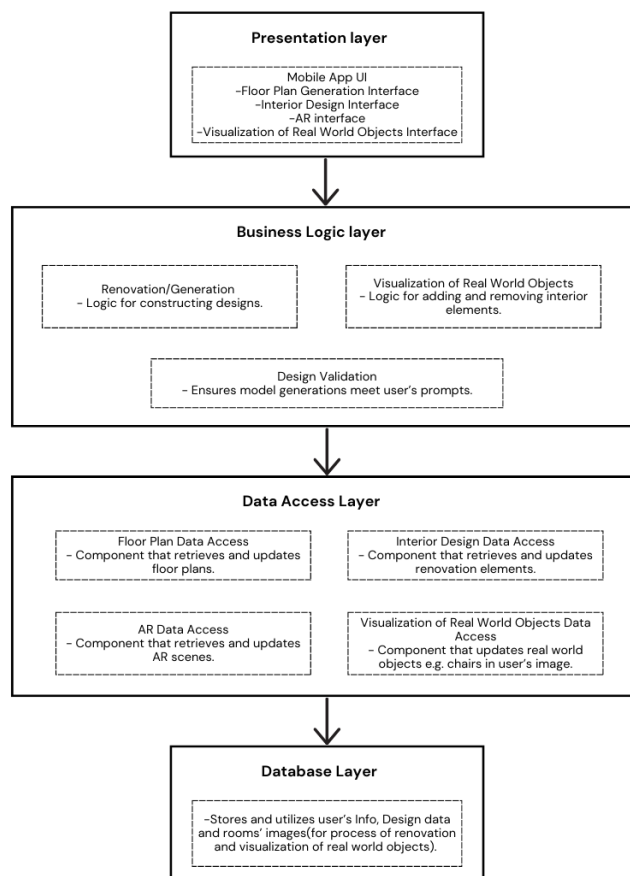


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagram

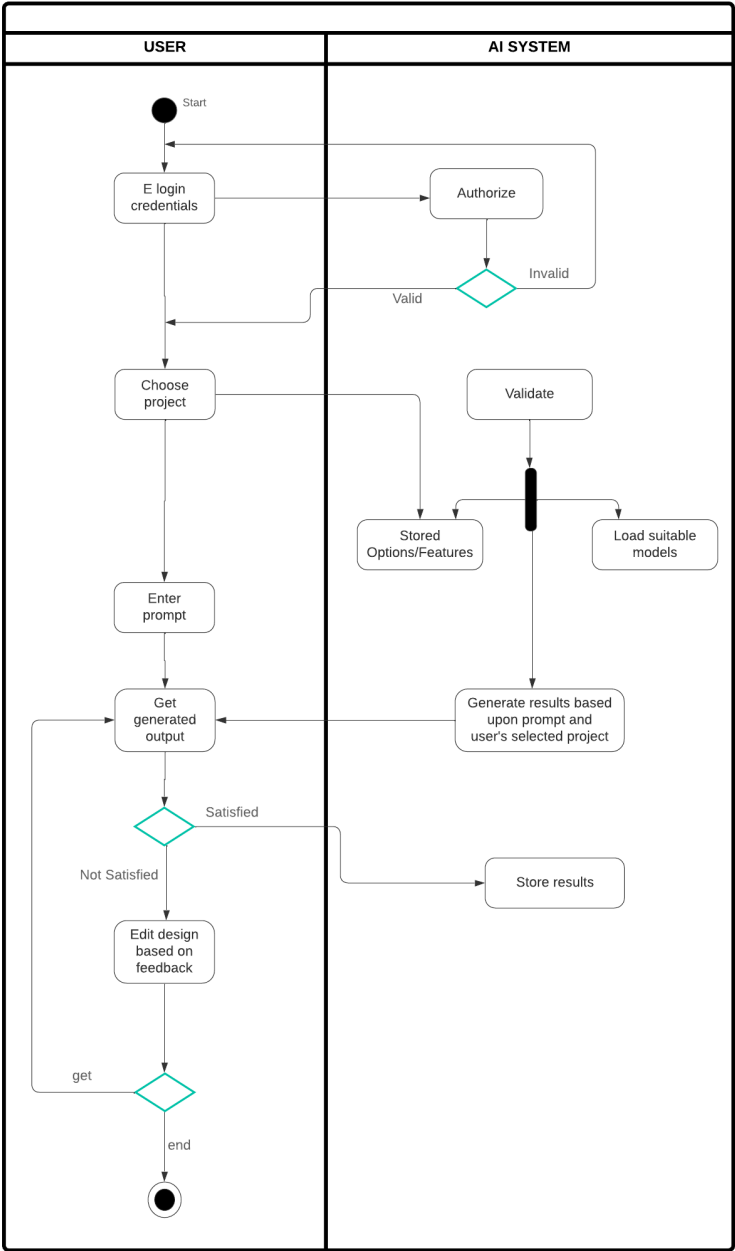


Figure 3.2: Activity Diagram

3.2.2 Data Flow Diagram

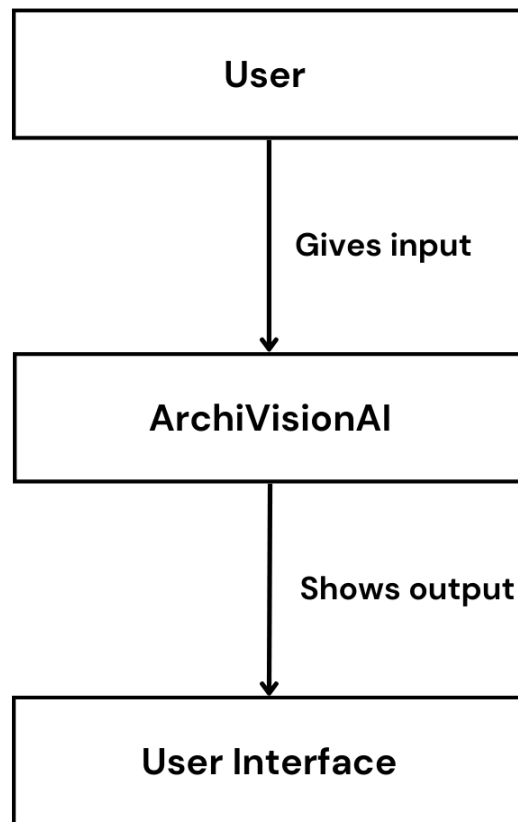


Figure 3.3: Data Flow Diagram - Level 0

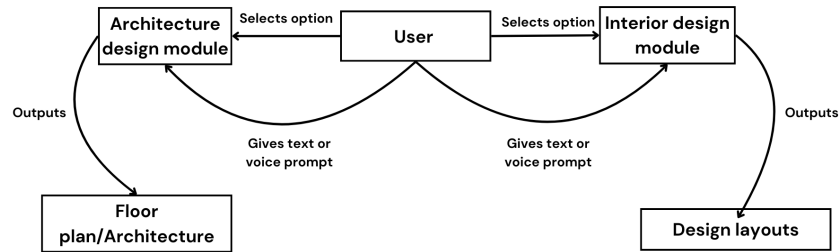


Figure 3.4: Data Flow Diagram - Level 1

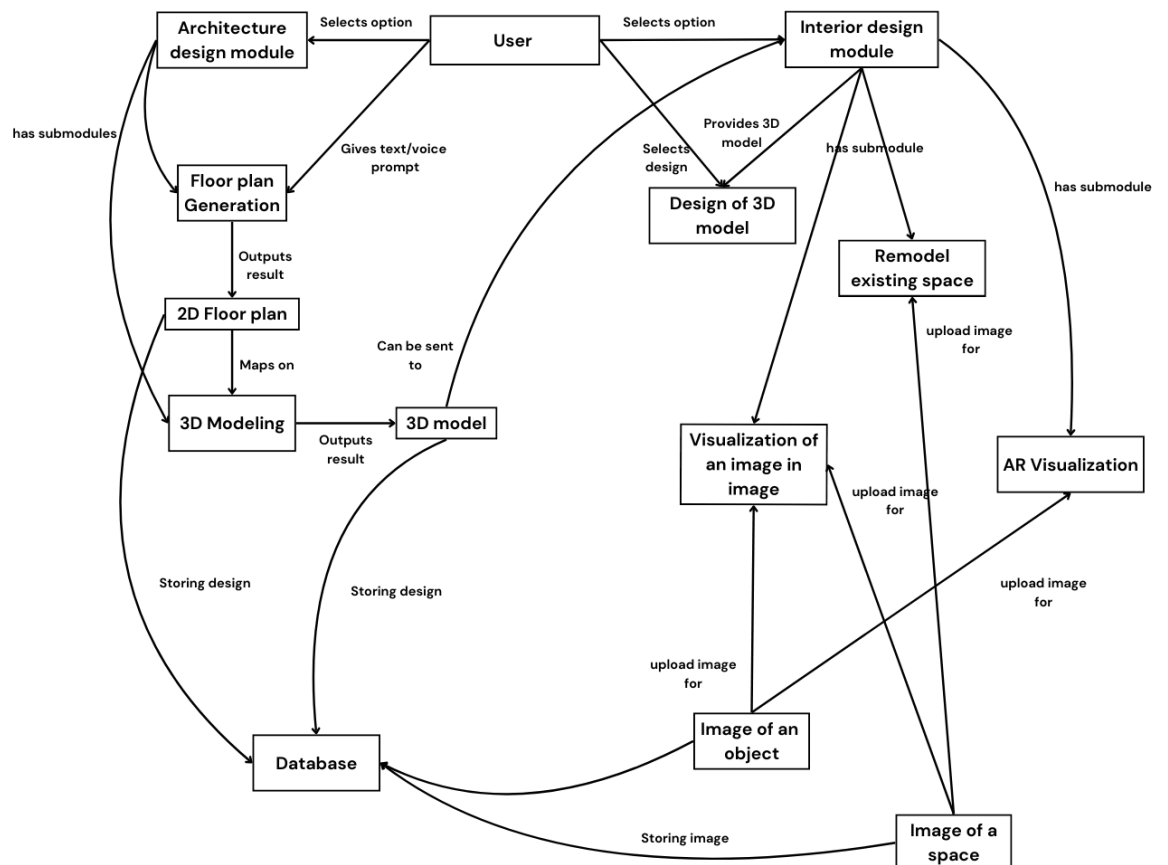


Figure 3.5: Data Flow Diagram - Level 2

3.2.3 State Level Sequence Diagram

3.2.3.1 Floor Plan and 3D Model

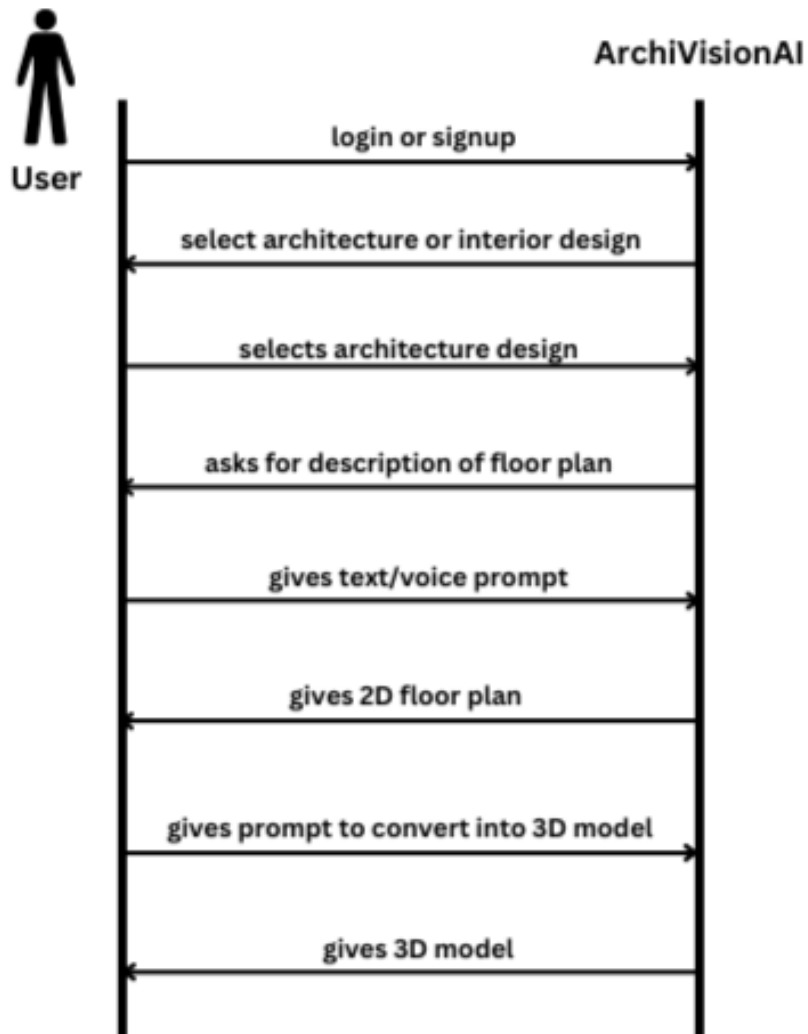


Figure 3.6: Floor plan and 3D model

3.2.3.2 Interior Design of 3D Model

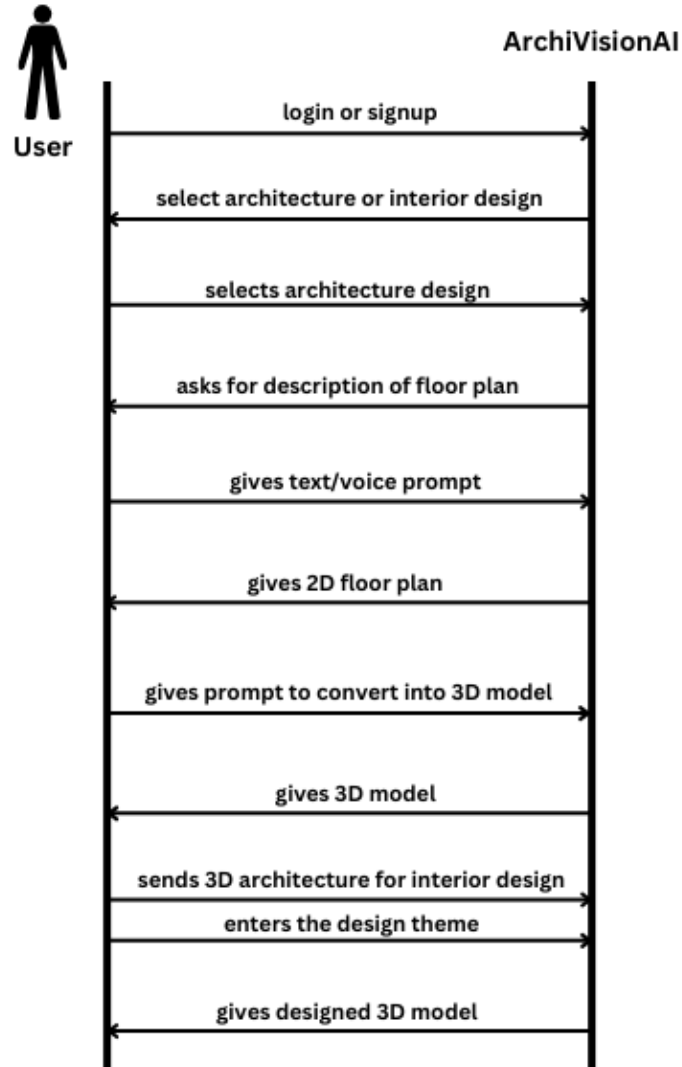


Figure 3.7: Interior Design of 3D Model

3.2.3.3 AR Feature

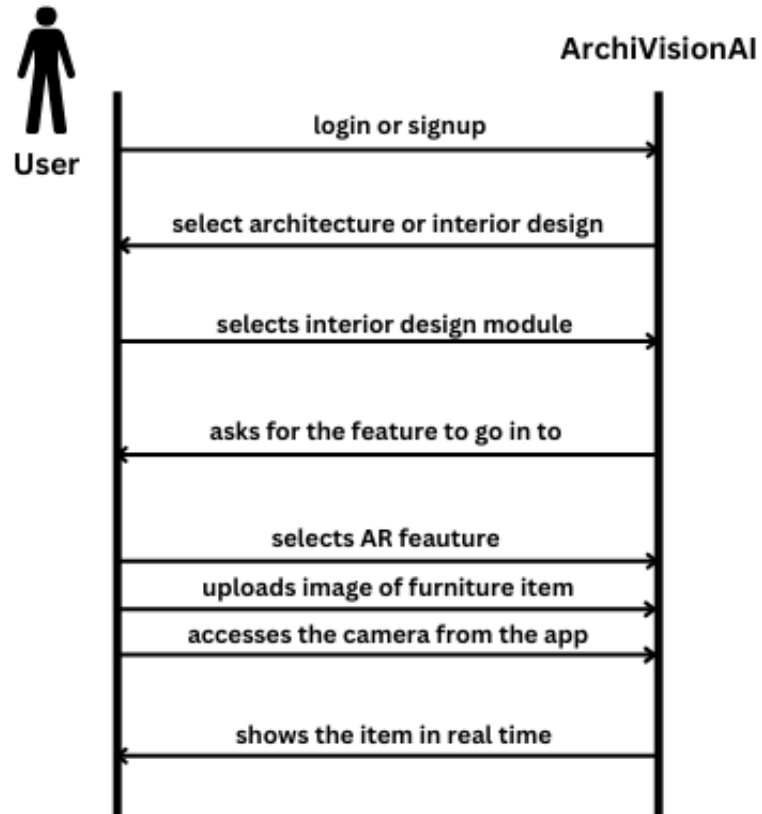


Figure 3.8: AR Feature

3.2.3.4 Remodeling of an Existing Space

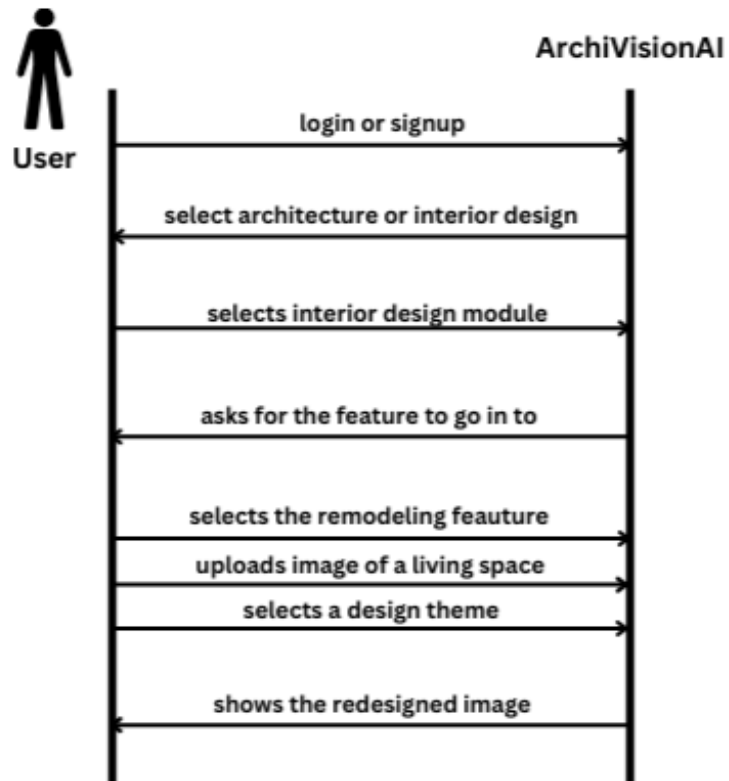


Figure 3.9: Remodeling of an existing space

3.2.3.5 Smart Purchase Preview

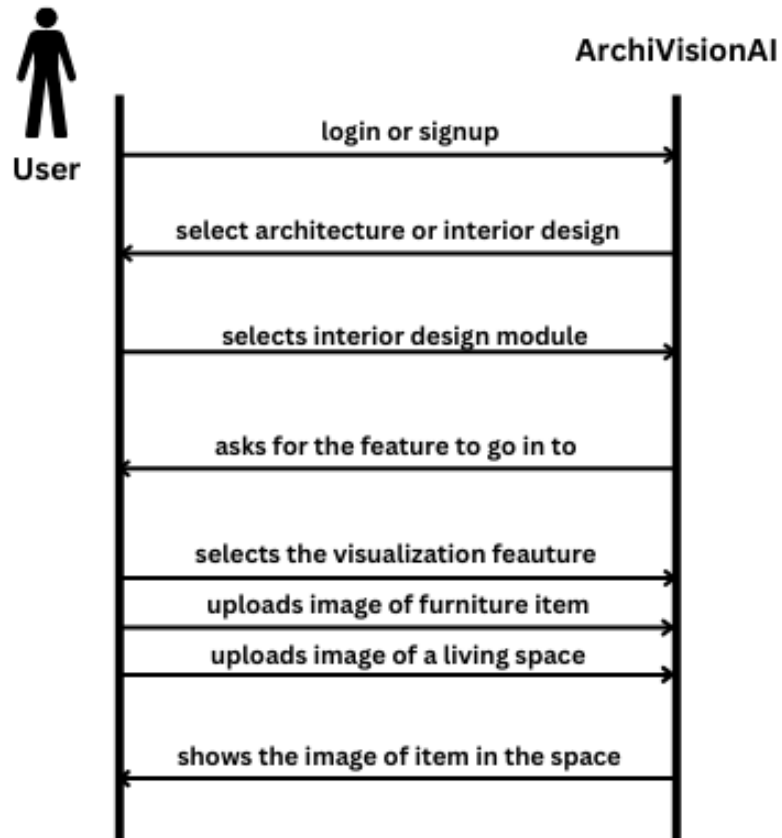


Figure 3.10: Smart Purchase Preview

3.2.4 State Transition Diagram

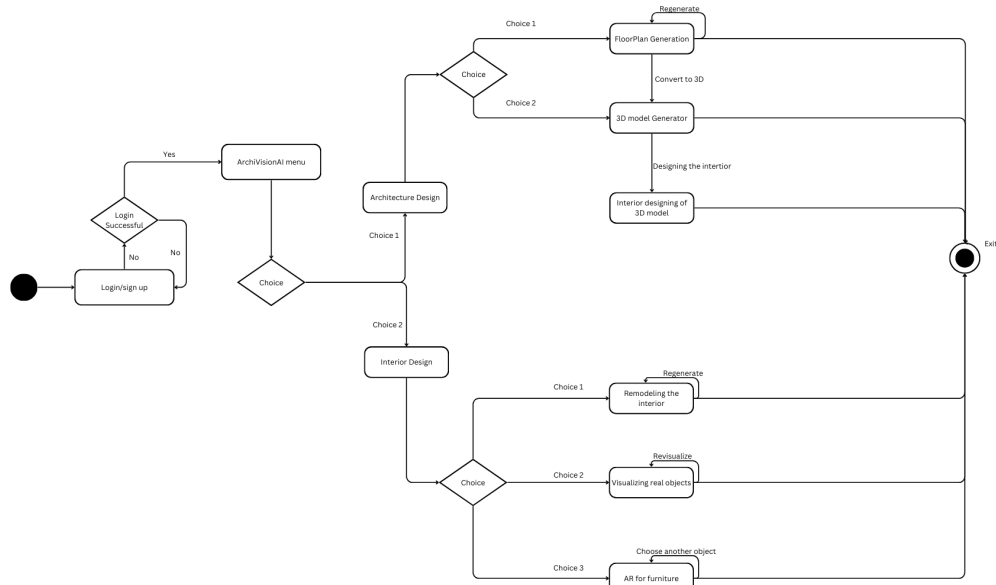


Figure 3.11: State Transition Diagram

3.3 Data Design

The information domain of our system is transformed into structured data that is suitable for the implementation of different functionalities including architectural, interior design, augmented reality (AR) and so on. The data is stored and processed in a such a way that these features are implemented efficiently.

3.3.1 Data Entities and Structures

The major data entities as follows:

1. User Data:

- **Storage:** To be stored in a database with tables for users and authentication credentials.
- **Data Fields:** UserID, Username, Email, Password and Project files.
- **Processing:** User data is retrieved for authentication in login/sign-up part. Project files are stored for reuse and modifications.

2. Floor Plans Data:

- **Storage:** Floor plans are stored as files (2D and 3D models).
- **Data Fields:** FloorPlanID, FilePath.
- **Processing:** The user inputs the prompts regarding the generation of floorplans. This is saved for future use such as conversion to 3D model, designing interior of 3D model and so on.

3. Interior Design Data:

- **Storage:** Interior designs generated by AI are stored as files (2D and 3D),
- **Data Fields:** DesignID, FilePath.
- **Processing:** The users inputs the prompts regarding the styling of 2D or 3D spaces and can store it for future reference.

4. AR Data:

- **Storage:** Real-time data captured by AR devices is stored temporarily in memory or on-device storage.
- **Data Fields:** ObjectID, ARPlacementData.
- **Processing:** Real-time visualization allows users to overlay virtual furniture on their real-world space. This information is deleted when the session ends.

3.3.2 Databases or Data Storage

The key databases and data storage items used are:

1. **User Authentication Database:** Stores all user credentials such as username ,password and project files.
2. **Database for interior, architecture design and AR:** Holds the 2D and 3D floor plans generated by users along with interior design files.AR information will be temporarily stored. Mostly cloud storage will be used.

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